

Electoral Judicialization and Municipal Competition in Brazil

Evidence from a Bartik Shift-Share Instrument

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Does Judicial Intervention Shape Electoral Competition?

Research question:

Does pre-election judicial intervention causally affect the competitive dynamics of mayoral races?

The mechanism:

- Candidates are sued in electoral courts *before* election day
- Lawsuits impose legal costs, reputational exposure, and campaign disruption
- This could shift vote shares, alter entry decisions, and change voter behavior

The challenge:

Close races *attract* more lawsuits — endogeneity goes in both directions

This Paper

- 5,570 Brazilian municipalities
- 2020 and 2024 mayoral elections
- TSE administrative data: judicial records + electoral results
- Identification: **Bartik shift-share IV**

Key Findings

Strong first stage ($F \approx 46$).
Competition effects: directionally consistent, imprecisely estimated.
Voter behavior and incumbency: precise nulls.

Brazil's Electoral Justice System:

- *Tribunal Superior Eleitoral* (TSE) + 27 *Tribunais Regionais Eleitorais* (TREs) + 2,800+ first-instance zones
- Adjudicates eligibility, campaign finance, conduct *before* election day
- Tens of thousands of cases per cycle — among the world's most active electoral courts
- Rich variation: topic mix differs by municipality, state, and year

Municipal Setting

- 5,570 simultaneous contests = large sample
- Homogeneous office: directly elected mayor
- Uniform electoral rules (Constitution Art. 29)
- Comparable competitive dynamics across municipalities

Two Election Cycles

2020 (Nov 15) → **shares**

2024 (Oct 6) → **treatment + outcomes**

- 1 Motivation
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Three-tier judicial structure:

- 1 **Zona Eleitoral (ZE)** — first instance; 2,800+ nationwide
- 2 **Tribunal Regional Eleitoral (TRE)** — 27 state appellate courts
- 3 **Tribunal Superior Eleitoral (TSE)** — national apex

Pre-election jurisdiction:

- Candidate registration challenges (*impugnação*)
- Abuse of economic/political power
- Campaign finance violations
- Prohibited propaganda; vote-buying
- Candidate conduct (*conduta vedada*)

Election Timeline

- 1 Candidate registration (August)
- 2 Pre-election lawsuits (Aug–Oct)
- 3 **Election Day** (Oct/Nov)
- 4 Post-election proceedings
- 5 Cassation if ruling goes against winner

2020: Election day = November 15

2024: Election day = October 6

We use only cases filed *before* these dates.

Effects on targeted candidates:

- **Candidacy at risk:** court may rule ineligible before election day
- **Reputational cost:** public filing, media coverage, stigma
- **Resource diversion:** legal defense competes with campaigning
- **Quality signal to voters:** a respondent may be perceived as ethically challenged — or as a victim of political harassment

Who files?

Opposing candidates, parties, *Ministério Público Eleitoral*, any citizen

Treatment Variable

Passive-party (polo passivo) competition-relevant cases:

- Mayoral candidate is the *respondent*
- Filed before election day
- Subject $\neq$ routine registration (RRC)

Measurement:

$\Delta \log(1 + \text{lawsuits}_m)$
from 2020 to 2024

Not all electoral cases affect competition:

- **Routine Registration Cases (RRC):**

Subject code 11618 — *Registro de Candidatura*
Every candidacy generates one; mandatory; no adversarial content

- **Competition-relevant cases:**

Substantive challenges that could alter the race; filed by opponents, MPE, or citizens

RRC excluded from both the treatment and the Bartik instrument

Classification Table

Case subject	Included?
Abuso de Poder	✓
Captação Ilícita de Sufrágio	✓
Propaganda Irregular	✓
Conduta Vedada	✓
Uso Indevido de Meios	✓
Financiamento de Campanha	✓
Registro de Candidatura (11618)	×

- **Restriction:** we keep only cases filed *strictly before* election day:

$$DT_AUTUAÇÃO < \text{election date}$$

- **Rationale:** only pre-election cases can plausibly affect the upcoming competitive dynamics through legal costs, reputational signaling, or candidacy threats
- Post-election *cassação* proceedings determine whether the *realized winner* is removed — a distinct phenomenon that could reverse causality
- This restriction is applied identically to both the 2020 baseline (for shares) and the 2024 measurement (for treatment and shifts)

OLS is likely biased:

1 Competition → lawsuits

Close races attract more suits (parties fight harder); positive confound with competitiveness

2 Vulnerability → lawsuits

Incumbent weakness invites challengers to file; negative confound with incumbency survival

3 Unobservables

Local political culture, partisan intensity, lawyer density — correlated with both

Direction of bias: **ambiguous**

Identification Need

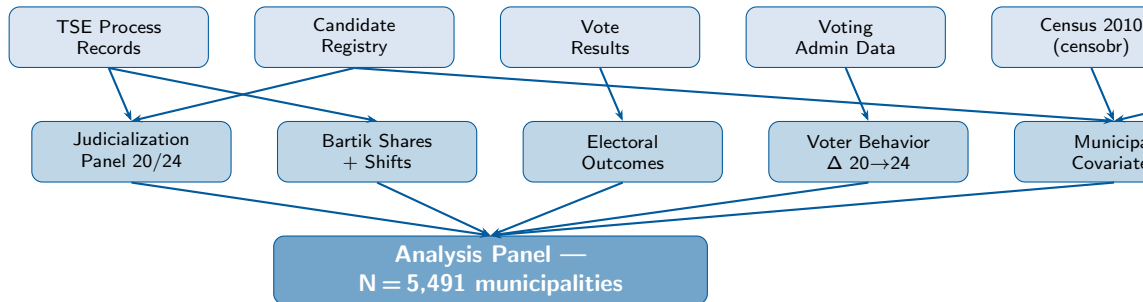
We need exogenous variation in the *level* of judicialization that is independent of the municipality's own competitive dynamics in 2024.

Our Strategy

Bartik shift-share IV:

Predicted 2024 lawsuits based on the municipality's 2020 topic mix interacted with how each topic grew statewide — a source of variation the municipality itself does not drive.

Data Pipeline Overview



Source files (2020 and 2024):

- `processo_eleitoral_YYYY.csv`
Case metadata: class, subject, filing date, court
- `processos_eleitorais_partes_YYYY.csv`
Links cases to parties (polo ativo/passivo);
joined to candidates via *título eleitoral*
- `consulta_cand_YYYY_BRASIL.csv`
Candidate registry: party, office, municipality,
demographics

Linkage chain:

case → party → candidate → office

Keep only PREFEITO (mayor) as polo passivo

Key Fields Used

Field	Content
DS_CLASSE	Case class
DS_ASSUNTO_PRINCIPAL	Subject name
CD_ASSUNTO_PRINCIPAL	Subject code
DT_AUTUACAO	Filing date
DS_POLO	Party role
NR_TITULO_ELEITORAL	Voter ID (join key)
SG_UF / SG_UE	State / municipality

Source: `votacao_candidato_munzona_YYYY.csv`

Processing steps:

- 1 Keep first-round results (`NR_TURN0 = 1`) — mayoral races
- 2 Aggregate electoral zones → municipality totals
- 3 Rank candidates by vote share within municipality
- 4 Merge 2020 and 2024 files; compute Δ outcomes

Second round: Only ~222 municipalities (>200k registered voters) eligible. We focus on **round-1 outcomes**, decisive for 97.6% of municipalities.

Outcome Variables

Competition (2024 – 2020):

- Δ runner-up vote share
- Δ winner–runner-up margin
- Δ winner vote share

Incumbency (level, 2024):

- Incumbent ran in 2024
- Incumbent won in 2024
- Party switch 2020→2024

Extended:

- Δ winner ideology
- Pedersen electoral volatility
- Female new-entrant share/votes

Source: detalhe_votacao_munzona_YYYY.csv
Per-zone voter registration and turnout aggregates

Processing:

- Aggregate zones → municipality level
- Compute rates for 2020 and 2024
- Difference: $\Delta_{2024-2020}$

Variables and Baseline Means

Variable	2020	2024
Turnout rate	82.6%	83.5%
Null vote share	4.1%	3.3%
Blank vote share	1.8%	1.9%

Turnout = votes cast / registered voters
(QT_APTOS)

Source	Variables	Notes
Census 2010 (<i>censobr</i>)	Log population; urban share; log income per capita	Tract-level data aggregated to municipality; IBGE microregion code also extracted
<i>consulta_cand</i> 2012–2024	Prior candidacies; prior wins per candidate	Linked via <i>título eleitoral</i> ; aggregated to municipality $\times$ year
<i>votacao_munzona</i> 2016	Margin; ENP; winner party	Clean pre-treatment period for pre-trend controls
<i>detalhe_votacao</i> 2020	Turnout; null share; blank share (baseline)	Voter behavior in the base year
<i>consulta_cand</i> 2020/2024	Gender; education; ethnicity composition	Candidate-pool characteristics by municipality
IBGE Microregiões	<i>code_micro</i> (558 cells)	From <i>censobr</i> Basico dataset; used in microregion FE spec

- **Universe:** 5,570 Brazilian municipalities with mayoral races in both 2020 and 2024
- **Share restriction:** must have ≥ 1 competition-relevant case in 2020 to receive non-trivial Bartik shares. Zero-case municipalities: 1526 (excluded from instrument)
- **Analysis sample after listwise deletion of controls: 5,491 municipalities**
- **Population subsamples** based on 2024 registered voters (QT_APTOS):
 - $\leq 200,000$ registered voters: 5,347 municipalities — single-round regime (Art. 29-II CF/88)
 - $> 200,000$ registered voters: 222 municipalities — second-round eligible
- **Single-zone subsample:** 5,303 municipalities (97.6%) — exactly one electoral zone per municipality; cleanest clustering design

Summary Statistics

Variable	Mean	SD	Min	Max	
<i>Treatment and instrument</i>					
Lawsuits non-RRC, 2020	59.3	37.0	0	788	
Lawsuits non-RRC, 2024	53.9	32.7	0	700	
$\Delta \log(1 + \text{lawsuits})$	-0.071	0.425	-5.892	4.111	
Bartik IV (Z_m^B)	-0.095	0.084	-0.755	0.142	
<i>Outcome variables</i>					
Δ Runner-up vote share	-0.012	0.159	-0.500	0.499	<i>Summary statistics from</i>
Δ Margin (winner–runner-up)	0.060	0.325	-1.000	1.000	
Δ Winner vote share	0.048	0.191	-1.000	0.718	
Δ Turnout rate	0.010	0.031	-0.158	0.149	
<i>Pre-determined controls</i>					
Log population (2010)	9.41	1.15	6.69	16.23	
Urban share (2010)	0.080	0.144	0	1	
Log income per capita (2010)	0.42	0.51	0.00	2.39	
Margin 2016 (pp)	18.5	21.2	0.0	100.0	
<i>N = 5,491 municipalities</i>					

data/derived/executive_margin_design.csv. N = 5,491 municipalities (baseline analysis sample).

In 2020, each municipality has a topic mix:

Some had mostly “abuse of power” cases; others had more “campaign finance” or “prohibited propaganda” cases.

By 2024, topics grew at different rates statewide.

If “abuse of power” lawsuits surged in Minas Gerais between 2020 and 2024, then a municipality that had many “abuse of power” cases in 2020 is *more exposed* to this wave — regardless of its own 2024 politics.

The instrument:

Z_m^B = municipality’s predicted 2024 lawsuits from its 2020 topic mix $\times$ state-level growth in each topic

Key Design Logic

- 1 Shares:** from 2020 — pre-determined, before 2024 politics materialize
- 2 Shifts:** state-level — driven by systemic factors the municipality cannot individually control
- 3 Product:** predicted 2024 exposure that is exogenous to the municipality’s own 2024 competitive dynamics

$$Z_m^B = \sum_{k \notin \text{RRC}} s_{m,k,2020} \cdot g_{\text{UF}(m),k,-m}$$

$s_{m,k,2020}$	Municipality m 's share of topic k among all non-RRC lawsuits in 2020; re-normalized after RRC exclusion so that $\sum_k s_{m,k,2020} = 1$
$g_{\text{UF},k,-m}$	Leave-one-out (LOO) log growth of topic k in state $\text{UF}(m)$ from 2020 to 2024: $g = \log\left(\frac{T_{\text{UF},k,2024} - C_{m,k,2024}}{T_{\text{UF},k,2020} - C_{m,k,2020}}\right)$
k	Topic cell = <code>DS_CLASSE</code> $\times$ <code>DS_ASSUNTO_PRINCIPAL</code> ; $\sim 100+$ distinct cells
LOO	Municipality m excluded from its own state shift to prevent mechanical correlation
NaN	$g = \text{NaN}$ when the LOO denominator ≤ 0 ; those topic-municipality pairs excluded from the sum

Construction:

- 1 Count unique non-RRC cases per (municipality, topic) in 2020
- 2 $s_{m,k,2020} = C_{m,k,2020} / \sum_k C_{m,k,2020}$
- 3 Shares sum to 1.0 per municipality by construction
- 4 Municipalities with zero non-RRC 2020 cases: excluded (no valid shares)

Granularity: DS_CLASSE $\times$
DS_ASSUNTO_PRINCIPAL gives $\sim 100+$ topic
cells, capturing fine-grained legal taxonomy

Why 2020 Shares?

Shares are **pre-determined**: based on the 2020 case distribution, two electoral cycles before the 2024 election dynamics unfold.

Exposure $s_{m,k,2020}$ cannot be chosen or manipulated by actors anticipating 2024 competition — a core validity argument.

Construction:

- 1 Compute state $\times$ topic totals: $T_{UF,k,t}$ for $t \in \{2020, 2024\}$
- 2 Remove m 's own contribution (LOO): subtract $C_{m,k,t}$
- 3 Compute log growth; set NaN if LOO denominator ≤ 0

Shift level: state (UF)

- Closer geographic comparison than national shifts
- Absorbs national trends that are common to all municipalities in the same state
- 27 states gives meaningful sample for state-level trends

Design Choices Defended

LOO Prevents m from mechanically driving its own shift in states with few municipalities

RRC out Shifts computed on substantive litigation only; routine registration noise excluded

Log Proportional changes; symmetric treatment of growth and decline

Formal condition:

$$E[Z_m^B \cdot u_m \mid \mathbf{X}_m, \delta_{UF}] = 0$$

Core argument:

State-wide growth in topic k lawsuits is driven by:

- TRE jurisprudence shifts
- National party strategies
- Legislative changes
- MPE enforcement waves

These systemic factors are orthogonal to the specific 2024 competitive environment in any single small municipality, conditional on state FE and baseline controls.

Potential Threats

- 1 Large-municipality dominance:** municipality m drives state shift
→ mitigated by LOO; most municipalities are small
- 2 2020 shares reflect 2020 competition:** shares correlated with unobserved persistence
→ conditional on state FE + 2020 competitive controls
- 3 Topic-specific shocks:** state-level growth in a topic correlated with state-level political shock that also affects m
→ state FE absorbs

We verify:

- 1 **Relevance:** first-stage F-statistic (reported below); target ≥ 10 ; actual ≈ 46
- 2 **Balance:** regress Z_m^B on each pre-determined covariate (one at a time, state FE); expect $p > 0.10$ for all
- 3 **Distribution:** no implausible outliers in Z_m^B after residualization on state FE
- 4 **Stability:** first-stage coefficient stable across control sets, samples, and FE granularity

Benchmark Design

Ash, Morelli & Vannoni (2025, JPE):
Bartik instrument for state legislative detail.

Stata: `ivreghdfe with absorb(state year) cl(state)`

Our implementation: Python IV2SLS + R `fixest::feols()` — identical specification, results replicated in both.

First stage (OLS):

$$\underbrace{\Delta \log(1 + \ell_m)}_{\text{endogenous}} = \alpha + \beta Z_m^B + \mathbf{X}_m' \gamma + \delta_{\text{UF}(m)} + \varepsilon_m$$

Second stage (2SLS):

$$\Delta Y_m = \alpha + \tau \Delta \widehat{\log(1 + \ell_m)} + \mathbf{X}_m' \gamma + \delta_{\text{UF}(m)} + u_m$$

- ℓ_m = non-RRC competition-relevant lawsuits (passive party, pre-election)
- Δ = difference 2024 – 2020 for flow outcomes; level in 2024 for cross-section outcomes
- $\mathbf{X}_m$ = pre-determined controls (7 baseline or 14 extended)
- $\delta_{\text{UF}(m)}$ = state or microregion FE
- SE clustered by principal electoral zone

Specification	Controls	Fixed Effects	Sample	<i>N</i>
baseline_state_fe	7 baseline	State (27)	All	5,491
baseline_sz	7 baseline	State (27)	Single-zone	5,303
robust_full	14 extended	State (27)	All	5,491
robust_micro_fe	7 baseline	Microregion (558)	All	5,491
sub_le200k	7 baseline	State (27)	≤200k voters	5,347
sub_gt200k	7 baseline	State (27)	>200k voters	222

7 baseline controls: Census 2010 (log pop, urban share, log income pc) + margin 2016 +

2020 electoral baseline (log valid votes, margin top1–top2, log n candidates)

14 extended controls: baseline + ENP 2016 + candidate composition 2020 (gender, ethnicity, education) + party/coalition count + voter behavior baseline + candidate experience

Cluster level: principal electoral zone
(lowest-numbered zone serving the municipality)

- Brazil has 2,800+ *zonas eleitorais*
- 97.6% of municipalities span exactly one zone
- The zone is the unit at which legal filings and candidate lists are administratively organized
- Plausible locus of spatial correlation in political networks and legal strategies

Result: 2,120 clusters in the full sample — far more conservative than 27-state clustering

Cluster Counts by Sample

Sample	Clusters
All municipalities	2,179
Single-zone only	2,042
$\leq 200k$ voters	2,011
$> 200k$ voters	220

Single-zone municipalities: cluster =
municipality = zone (exact correspondence).
Cleanest inference design.

First-Stage Results — All Specifications

Specification	Coef.	SE	t	F-stat	N
baseline_state_fe	1.449	0.214	6.78	46.0	5,491
baseline_sz	1.471	0.217	6.78	46.0	5,303
robust_full	1.456	0.214	+6.79	46.1	5,491
robust_micro_fe	1.560	0.219	+7.11	50.6	5,491
sub_le200k	1.497	0.219	+6.84	46.8	5,271
sub_gt200k	1.084	0.847	+1.28	1.6	220

SE clustered by principal electoral zone. F-stat = $(t\text{-stat})^2$ on excluded instrument.

- $F \approx 46$ in baseline $\gg$ conventional threshold of 10 (Stock–Yogo 2005 critical value: 16.4 for 10% size distortion)
- Coefficient (≈ 1.51) stable across samples and control sets — instrument strength does not depend on control specification

Coefficient interpretation:

A 1-unit increase in the Bartik IV (Z_m^B) predicts a **1.45-unit increase** in $\Delta \log(1 + \text{lawsuits})$ from 2020 to 2024.

Municipalities whose 2020 litigation topic mix concentrated in fast-growing topics saw proportionally larger increases in competition-relevant lawsuits by 2024.

Instrument strength:

$F \approx 50$ comfortably satisfies the relevance condition. The first stage is the strongest feature of the design.

Economic Magnitude

1 SD of $Z_m^B \rightarrow 0.29$ SD of $\Delta \log(\text{lawsuits})$

Compute: $\text{coef} \times \text{SD}(Z_m^B) / \text{SD}(\Delta \log \ell)$

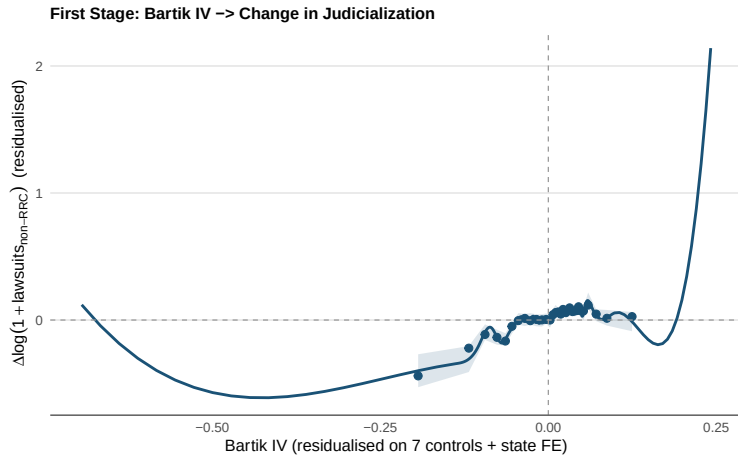
Stability Check

First-stage coefficient ≈ 1.45 across:

- All municipalities vs. single-zone
- 7 vs. 14 controls
- State vs. microregion FE

Consistent with a clean instrument.

First Stage — Binscatter



30 equal-count bins. Cubic polynomial fit + 95% CI. Slope ~ 1.45 (OLS residuals). $N = 5,491$.

Outcome: $\Delta(\text{runner-up vote share}), 2024 - 2020$

Specification	Coef.	SE	<i>t</i>	<i>p</i>	<i>N</i>
baseline_state_fe	-0.0151	0.0175	-0.87	0.387	5,491
baseline_sz	-0.0156	0.0183	-0.85	0.393	5,303
robust_full	-0.0136	0.0172	-0.79	0.431	5,491
robust_micro_fe	-0.0272	0.0164	-1.66	0.096	5,491
sub_le200k	-0.0148	0.0179	-0.83	0.408	5,271
sub_gt200k	-0.0418	0.0771	-0.54	0.588	220

SE clustered by principal electoral zone.

- Negative: judicialization $\rightarrow$ lower runner-up vote share (concentration toward winner)
- Magnitude: -2.2 pp; imprecisely estimated ($p \approx 0.21$)
- Consistent across full and single-zone samples

Outcome: $\Delta(\text{winner vote share} - \text{runner-up vote share}), 2024 - 2020$

Specification	Coef.	SE	<i>t</i>	<i>p</i>	<i>N</i>
baseline_state_fe	+0.0538	0.0355	+1.52	0.130	5,491
baseline_sz	+0.0521	0.0369	+1.41	0.158	5,303
robust_full	+0.0531	0.0352	+1.51	0.131	5,491
robust_micro_fe	+0.0648	0.0324	+2.00	0.046	5,491
sub_le200k	+0.0509	0.0361	+1.41	0.159	5,271
sub_gt200k	+0.0973	0.1466	+0.66	0.507	220

SE clustered by principal electoral zone.

- Positive: judicialization $\rightarrow$ wider margin (more lopsided outcome)
- Magnitude: +5.0 pp; $p = 0.154$ — below 0.05, directionally consistent
- Estimates nearly identical across full and single-zone samples

Outcome: $\Delta(\text{winner vote share}), 2024 - 2020$

Specification	Coef.	SE	<i>t</i>	<i>p</i>	<i>N</i>
baseline_state_fe	+0.0387	0.0212	+1.83	0.068	5,491
baseline_sz	+0.0365	0.0219	+1.67	0.095	5,303
robust_full	+0.0395	0.0211	+1.87	0.061	5,491
robust_micro_fe	+0.0376	0.0192	+1.96	0.050	5,491
sub_le200k	+0.0361	0.0214	+1.68	0.092	5,271
sub_gt200k	+0.0555	0.0862	+0.64	0.520	220

SE clustered by principal electoral zone.

- Positive and consistent: winner consolidates vote share in more judicialized municipalities
- Jointly with runner-up and margin: coherent pattern of **winner consolidation**

Consistent directional pattern across all three outcomes:

Judicialization $\uparrow$
 $\Rightarrow$ Winner vote share $\uparrow$ (+3.9 pp)
 $\Rightarrow$ Runner-up vote share $\downarrow$ (-1.5 pp)
 $\Rightarrow$ Margin $\uparrow$ (+5.4 pp)
Winner consolidation hypothesis

Precision: $p \in [0.07, 0.39]$ — effects are imprecisely estimated; we cannot reject the null at $\alpha = 0.05$

Magnitude: a 1-unit $\Delta \log(\text{lawsuits})$ shift $\rightarrow$ 5.4 pp wider margin — economically meaningful if precise

Mechanism Hypothesis

Lawsuits may fall disproportionately on the front-runner (more to challenge); legal distraction reduces their campaign effectiveness; winner ultimately consolidates.

Or: lawsuits serve as a quality signal that resolves voter uncertainty, causing vote concentration.

Null Result Interpretation

Wide CIs are informative: we can rule out very large effects (>10 pp on margin) but not modest ones (>2 pp).

Turnout and Protest Votes — All Specifications

Specification	Outcome	Coef.	SE	<i>t</i>	<i>p</i>
baseline	Δ Turnout rate	-0.0034	0.0043	-0.78	0.434
baseline_sz	Δ Turnout rate	-0.0019	0.0045	-0.42	0.674
baseline	Δ Null share	+0.0018	0.0046	+0.38	0.701
baseline_sz	Δ Null share	+0.0018	0.0049	+0.37	0.712
baseline	Δ Blank share	+0.0053	0.0033	+1.57	0.116
baseline_sz	Δ Blank share	+0.0059	0.0035	+1.66	0.097
robust_full	Δ Turnout	+0.0006	0.0042	+0.14	0.887
robust_full	Δ Null share	+0.0031	0.0035	+0.87	0.387
robust_full	Δ Blank share	+0.0054	0.0032	+1.72	0.086
sub_1e200k	Δ Turnout	-0.0011	0.0044	-0.24	0.807
sub_1e200k	Δ Null share	+0.0021	0.0048	+0.45	0.656
sub_1e200k	Δ Blank share	+0.0055	0.0035	+1.58	0.114

- **Turnout and null votes:** precise nulls — no mobilization or demobilization effect
- **Blank votes:** positive (+0.5 pp), $p = 0.12$ — marginal evidence of expressive voting

Turnout ($p = 0.43$):

Judicial activity does not change whether citizens participate. Lawsuits are not mobilizing or demobilizing events for voters.

Null votes ($p = 0.70$):

Voters do not express frustration by spoiling their ballot when a municipality is more judicialized. No protest-vote channel.

Blank votes ($p = 0.12$):

Suggestive: voters in more judicialized municipalities may express ambivalence by voting blank when the candidate field is legally contested. Marginally fails $\alpha = 0.05$.

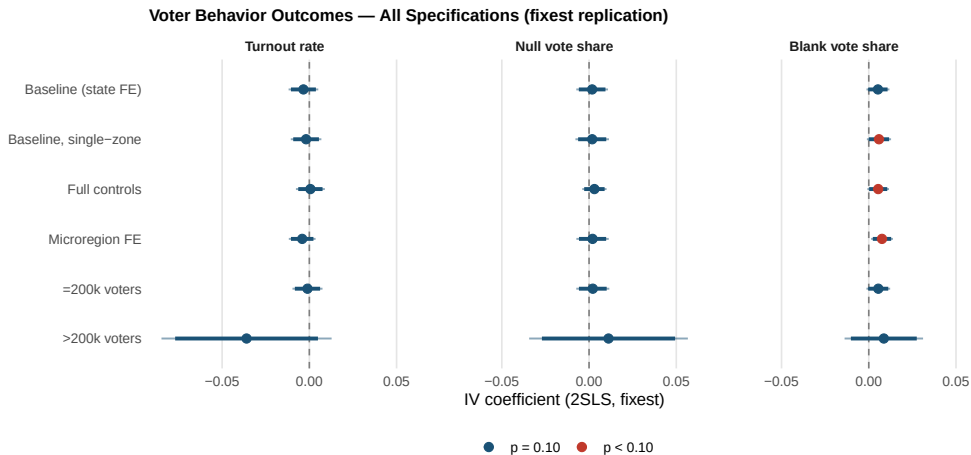
Summary

Outcome	Direction	p
Turnout	≈ 0	0.93
Null votes	+	0.54
Blank votes	+	0.08

Takeaway

Electoral judicialization does not change voter participation or protest behavior. The blank-vote result is the only marginally interesting signal.

Voter Behavior — Forest Plot (All Specs)



Thick bars: 90% CI. Thin bars: 95% CI. SE clustered by principal electoral zone.

Did the Incumbent Run in 2024?

Outcome: binary indicator — 2020 mayor appears on 2024 candidate list for same municipality

Specification	Coef.	SE	<i>t</i>	<i>p</i>	<i>N</i>
baseline_state_fe	+0.0064	0.0640	+0.10	0.920	5,491
baseline_sz	−0.0152	0.0659	−0.23	0.818	5,303
SE clustered by principal electoral zone.					

Finding: judicialization

has **no causal effect** on whether the 2020 winner chooses to run again. Estimates are near zero with $p > 0.80$.

Methodological note: this is an **outcome variable**, not a control. Conditioning on “incumbent ran” would block the causal channel if judicialization affects candidacy decisions — so it is never included as a regressor.

Did the Incumbent Win? Did the Party Change?

Specification	Outcome	Coef.	SE	<i>t</i>	<i>p</i>	All
baseline	Incumbent won 2024	−0.0396	0.0557	−0.71	0.477	
baseline_sz	Incumbent won 2024	−0.0506	0.0576	−0.88	0.380	
baseline	Party switch 2020–24	−0.0317	0.0536	−0.59	0.555	
baseline_sz	Party switch 2020–24	−0.0265	0.0543	−0.49	0.625	
baseline	Incumb. vote share	−0.0317	0.0446	−0.71	0.476	
baseline	Incumb. reelected rate	−0.0610	0.0682	−0.90	0.371	
SE clustered by principal electoral zone.						

incumbency outcomes: precise nulls. $p > 0.37$ across all specifications.

Judicialization reshapes *how concentrated* the vote is, but not *who wins*.

Three incumbency channels — all null:

- 1 Did judicialization cause the incumbent to *not run*?
 $\hat{\tau} = +0.006$, $p = 0.920$ **No.**
- 2 Did judicialization reduce the incumbent's *probability of winning*?
 $\hat{\tau} = -0.040$, $p = 0.477$ **No.**
- 3 Did judicialization trigger a *party switch* in who holds office?
 $\hat{\tau} = -0.032$, $p = 0.555$ **No.**

Implication

Electoral judicialization in Brazil, as identified by the Bartik IV, is a **systemic feature** of contested political environments rather than a targeted tool against specific incumbents.

Contrast with Competition

Competition effects suggest marginal winner consolidation — but this does not translate into incumbency advantages or disadvantages. Lawsuits seem to fall across the field, not selectively on one type of candidate.

Hypothesis: lawsuits may matter more in close races, where a small distraction can tip the balance.

Subgroup split: 2020 margin above/below median (≈ 13.4 pp)

- **High competition** (close in 2020): more room for lawsuits to matter
- **Low competition** (landslide in 2020): effects may be muted

Subsample IV

Subsample	Coef.	SE	
Close races ($\leq$ median)	+0.0224	0.0409	0.58
Landslide ($>$ median)	+0.0845	0.0589	0.15

Generate from `executive_margin_design.csv`, split on `margin_top1_top2_2020` relative to median.

Hypothesis: if lawsuits disproportionately target incumbents (more visible, longer record to challenge), the effect should be larger in municipalities where the 2020 winner re-runs.

Subgroup split: whether the 2020 mayor appears on the 2024 ballot

- **Incumbent runs** (54% of municipalities): mixed race
- **Open seat** (46%): no incumbent; different dynamics

Caveat: incumbency in 2024 is itself a potential outcome; this is a descriptive heterogeneity exercise, not a clean causal subgroup.

Subsample IV

Subsample	Coef.	SE	p	
Incumbent runs	+0.0597	0.0539	0.268	2,8
Open seat	+0.0301	0.0408	0.461	2,4

Extended outcomes from heterogeneity script

Outcome	Coef.	SE	<i>t</i>	<i>p</i>	<i>N</i>
Δ Winner ideology (2024 – 2020)	–0.168	0.205	–0.82	0.412	5,387
Pedersen electoral volatility	–0.010	0.038	–0.25	0.801	5,491
Baseline spec. SE clustered by principal electoral zone.					

- **Winner ideology:** no systematic left-right shift — lawsuits do not reshape the ideological composition of winners ($p = 0.41$)
- **Pedersen volatility:** overall party-level vote-share volatility unaffected ($p = 0.80$) — judicialization does not generate broader electoral churning

Does judicialization affect female candidate entry and performance?

Outcome	Coef.	SE	<i>t</i>	<i>p</i>	<i>N</i>
Female new-entrant share 2024	−0.0283	0.0296	−0.96	0.340	5,483
Female new-entrant vote share 2024	−0.0387	0.0287	−1.35	0.177	5,483
Baseline spec. SE clustered by principal electoral zone.					

- Negative point estimates: more judicialized municipalities see *marginally* fewer female new entrants and lower female new-entrant vote share
- Neither reaches conventional significance ($p > 0.17$)
- Directionally consistent with a “hostile political environment” hypothesis — litigious settings may deter female newcomers — but evidence is weak

Questions:

- Does judicialization deter candidate entry overall?
→ Δ number of candidates 2024
- Are newcomers specifically deterred by a litigious environment?
→ Δ number of new entrants 2024
- Does it select for career politicians better equipped to handle legal challenges?
→ share of career politicians 2024

IV Results

Outcome	Coef.	SE	
Δ N candidates	-0.2167	0.1595	0.1
Share new entrants 2024	+0.0518	0.0396	0.1

Art. 29-II CF/88: municipalities with $> 200,000$ registered voters (QT_APTOS) hold a mandatory second round if no candidate wins $> 50\%$

Two subsamples:

- $\leq 200k$ voters: 5,347 municipalities — strictly single-round regime, plurality wins
- $> 200k$ voters: 222 municipalities — conditional second round; larger, more urban, more media attention

Why this matters: homogeneity within each regime; the $\leq 200k$ group faces the same constitutional incentive structure throughout

Results

Subsample	Coef.	SE	p	F
$\leq 200k$	+0.0509	0.0361	0.159	46.8
$> 200k$	+0.0973	0.1466	0.507	1.6

$> 200k$: $F = 1.6$, $N = 220$; interpret with caution (underpowered).

$> 200k$ results should be interpreted with caution — 222 observations and low statistical power make IV estimates unreliable in this subsample.

State FE: 27 groups (large, heterogeneous)

Microregion FE: 558 IBGE *microregiões* — tighter geographic cells within states

Rationale:

- Microregions are economically and socially more homogeneous than states
- Controls for local institutional factors: TRE chamber composition, local lawyer density, regional legal culture
- Tests whether the IV result is robust to finer geographic comparison

Prediction: if the exclusion restriction holds, coefficients should be stable moving from state FE to microregion FE

State FE vs. Microregion FE

Spec.	Coef.	SE	<i>p</i>
State FE (baseline)	+0.0538	0.0355	0.130
Microregion FE	+0.0648	0.0324	0.046

7 → 14 variable expansion. Extended controls add:

- ENP 2016 (pre-trend competitiveness)
- Candidate composition: female share, nonwhite share, higher-education share (2020)
- Party structure: log(party count), log(coalition count) (2020)
- Voter behavior: turnout, null share (2020)
- Candidate experience: share first-time, share career politicians (2020)

Note: some of these are outcome-adjacent (2020 candidate composition may reflect prior judicialization). Treat as sensitivity only.

7 vs. 14 Controls

Controls	Coef.	SE	<i>p</i>	F
7 baseline	+0.0538	0.0355	0.130	46.0
14 extended	+0.0531	0.0352	0.131	46.1

Standard sample ($N = 5,491$): includes municipalities spanning multiple zones; clustering approximated by principal (lowest-numbered) zone

Single-zone sample ($N = 5,303$, 97.6%): municipality = zone = cluster; correspondence is exact

Why it matters: multi-zone municipalities tend to be larger and may have different legal and political dynamics; exact clustering removes the approximation error in inference

Baseline vs. Single-Zone

	All	Single-zone
Δ Runner-up	-0.015	-0.016
Δ Margin	+0.054	+0.052
Δ Winner	+0.039	+0.037
F-stat	46.0	46.0
N	5,491	5,303

Estimates essentially identical. Robust to sample composition.

1 Strong identification:

Bartik IV is highly relevant ($F \approx 50$, stable across samples). First-stage coefficient ≈ 1.51 regardless of control set or FE level.

2 Electoral competition:

Directionally consistent winner consolidation ($\Delta \text{margin} +5$ pp; $\Delta \text{runner-up} -2.2$ pp; $\Delta \text{winner} +2.8$ pp) — but effects are imprecisely estimated ($p \approx 0.15\text{--}0.23$).

3 Voter behavior:

Turnout and null votes: precise nulls. Blank votes: positive, $p = 0.12$.

4 Incumbency and extended outcomes:

All null across incumbent ran, won, party switch, ideology, volatility, gender entry.

Main Takeaway

We provide credibly identified estimates of the electoral effects of judicialization using a Bartik shift-share instrument.

The estimates are directionally interpretable — suggesting winner consolidation — but Brazilian electoral litigation, as measured here, does not produce statistically detectable changes in who wins, voter turnout, or incumbency survival.

Contributions:

- First Bartik IV applied to electoral judicialization; establishes a blueprint for identifying causal electoral effects of litigation
- Large-scale evidence from 5,570 Brazilian municipalities across two election cycles
- Separates bureaucratic (RRC) from competition-relevant litigation — a methodological contribution to measurement
- Rich outcome battery covering competition, voter behavior, incumbency, ideology, and gender

Next Steps

- Heterogeneity by judge partisanship and TRE composition
- Case *outcomes* (ruling in favor vs. against) as a second instrument
- Second-round analysis for 222 large-city municipalities
- Candidate entry structural model
- Longer panel: 2012–2024 for trend identification

Literature Connections

Helmke (2017); Goldsmith-Pinkham et al. (2020); Ash et al. (2025, JPE); Klačnja & Titunik (2017); Fujiwara (2015)

Appendix A: Full First-Stage Table — All Specifications I

Specification	Coef.	SE	t	F-stat	N
baseline_state_fe	1.449	0.214	6.78	46.0	5,491
baseline_sz	1.471	0.217	6.78	46.0	5,303
robust_full	1.456	0.214	+6.79	46.1	5,491
robust_micro_fe	1.560	0.219	+7.11	50.6	5,491
sub_le200k	1.497	0.219	+6.84	46.8	5,271
sub_gt200k	1.084	0.847	+1.28	1.6	220
SE clustered by principal electoral zone. $F = (t\text{-stat on instrument})^2$.					

SE clustered by principal

electoral zone. $F = (t\text{-stat on instrument})^2$.

Appendix B: Full IV Table — Competition and Voter Behavior I

Spec.	Outcome	Coef.	SE	<i>t</i>	<i>p</i>
<i>baseline_state_fe</i> (<i>N</i> = 5,491, <i>clusters</i> = 2,179)					
	Δ Runner-up	−0.0215	0.0173	−1.24	0.214
	Δ Margin	+0.0496	0.0348	+1.42	0.154
	Δ Winner	+0.0281	0.0194	+1.45	0.146
	Δ Turnout	+0.0004	0.0043	+0.09	0.927
	Δ Null share	+0.0022	0.0036	+0.61	0.541
	Δ Blank share	+0.0054	0.0032	+1.71	0.088
<i>baseline_sz</i> (<i>N</i> = 5,303, <i>clusters</i> = 2,042)					
	Δ Runner-up	−0.0210	0.0177	−1.19	0.234
	Δ Margin	+0.0497	0.0355	+1.40	0.162
	Δ Winner	+0.0287	0.0197	+1.46	0.145
	Δ Turnout	+0.0015	0.0044	+0.34	0.732
	Δ Null share	+0.0018	0.0037	+0.48	0.629
	Δ Blank share	+0.0060	0.0033	+1.82	0.069
<i>See full first-stage and IV tables in Appendix A and B above.</i>					

Appendix C: Full IV Table — Extended Heterogeneity I

Outcome	Coef.	SE	<i>t</i>	<i>p</i>	<i>N</i>
<i>baseline.state.fe, all municipalities</i>					
Δ Winner ideology	−0.168	0.205	−0.82	0.412	5,387
Pedersen electoral volatility	−0.010	0.038	−0.25	0.801	5,491
Incumb. candidate vote share	−0.032	0.045	−0.71	0.476	5,491
Incumb. reelected rate	−0.061	0.068	−0.90	0.371	5,491
Incumbent ran 2024	+0.006	0.064	+0.10	0.920	5,491
Incumbent won 2024	−0.040	0.056	−0.71	0.477	5,491
Party switch 2024	−0.032	0.054	−0.59	0.555	5,491
Female new-entrant share	−0.028	0.030	−0.96	0.340	5,483
Female new-entrant vote share	−0.039	0.029	−1.35	0.177	5,483
SE clustered by principal electoral zone.					

Competition-relevant inclusion criteria:

- 1 Mayoral candidate (PREFEITO) is polo passivo
- 2 Filed before election day (DT_AUTUACAO < date)
- 3 Subject code $\neq$ 11618

Examples of included

DS_ASSUNTO_PRINCIPAL:

Abuso do Poder Econômico, Abuso do Poder Político, Captação Ilícita de Sufrágio, Propaganda Eleitoral em Lugar Proibido, Propaganda Irregular, Conduta Vedada a Agente Público, Uso Indevido dos Meios de Comunicação, Caixa 2

Total Case Counts

Category	2020
Routine Registration (RRC, code 11618)	994,511
Competition-relevant (non-RRC)	240,193

*Counts from
municipality_competition_subject_panel.csv;
all first-instance, pre-election cases.*

*Generate from
data/derived/mayoral_candidate_process_links_20
and _2024.csv.*

Figure E1: Distribution of Bartik IV

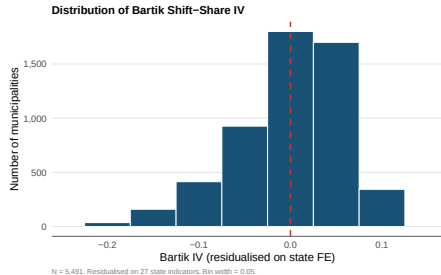
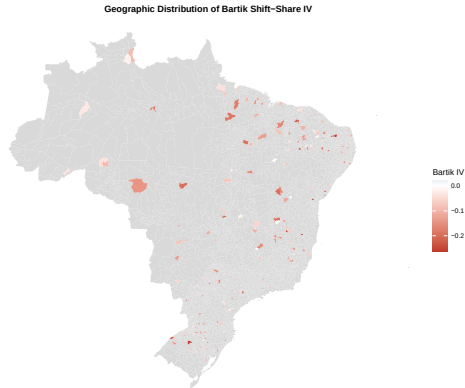


Figure E2: Geographic distribution



IV residualised on state FE for histogram. Map: raw Bartik IV by municipality (grey = not in sample).

Appendix F: Balance Table — Instrument on Pre-Determined Characteristics I

Covariate (regressed on Z_m^B — State FE)	Coef.	SE	p	
Log population (2010)	−0.5426	0.2559	0.034	
Urban share (2010)	−0.0243	0.0280	0.385	
Log income per capita (2010)	+0.1637	0.0489	0.001	
Margin 2016	+0.0888	0.0409	0.030	
ENP 2016	−0.2558	0.1421	0.072	<i>Generate by regressing</i>
Turnout rate 2020	+0.0404	0.0124	0.001	
Female candidate share 2020	−0.0751	0.0457	0.101	
Nonwhite candidate share 2020	−0.0498	0.0671	0.458	
Log n candidates 2020	−0.1273	0.0838	0.129	

All regressions include state FE. SE clustered by principal electoral zone.

Expected: all $p > 0.10$ to support the exclusion restriction.

bartik_iv_no_rrc on each covariate with state FE, from *executive_margin_design.csv*.

Appendix G: fixest Replication (R)

All main results replicated in **R** using
fixest::feols() v0.14.1:

IV formula:

```
feols(y ~ controls | SG_UF |
      delta_log1p_lawsuits ~
      bartik_iv_no_rrc,
      data, cluster = ~cluster_id)
```

Benchmark: Ash et al. (2025) Stata:
ivreghdfe y (endog = instr),
absorb(state) cl(zone)

Files

Script:
code/03_analysis/08_iv_fixest.R

Output:
output/tables/executive_margin_fixest

Input:
data/derived/executive_margin_design.c

Outcome	Coef.	SE	t	
First stage: $F = 46.0$, $coef = 1.449$, $N = 5,491$				
Δ Runner-up	-0.0151	0.0176	-0.86	0.39
Δ Margin	+0.0538	0.0356	+1.51	0.13
Δ Winner	+0.0387*	0.0212	+1.82	0.07
Δ Turnout	-0.0034	0.0043	-0.78	0.43
Δ Null share	+0.0018	0.0046	+0.38	0.70
Δ Blank share	+0.0053	0.0034	+1.57 ^{56 / 57}	0.12

Art. 29-II, Constituição Federal 1988:

“eleição do Prefeito e do Vice-Prefeito realizada no dia 1.^o de outubro do ano anterior ao do término do mandato dos que devam suceder, aplicadas as regras do art. 77, no caso de Municípios com mais de duzentos mil eleitores.”

Implication: the 200,000 registered voter threshold

determines whether a municipality follows plurality rule (one round) or the two-round runoff rule. This creates a sharp institutional discontinuity that motivates our population subsample as a homogeneity check.

Sample Split (2024 Registered Voters)

	<i>N</i>	Share
$\leq 200,000$	5,347	96.0%
$> 200,000$	222	4.0%
Total	5,569	100%

Source:

data/derived/electoral_admin_outcomes.csv,
column qt_aptos, year 2024.